CONSTRAINTS

Constraints were defined specifically for integrity purposes to ensure the accuracy, consistency, validity, and reliability of data within the Apex Digital Bank database. These constraints enforce business rules and prevent invalid or inconsistent data from being stored.

We have a number of constraints enforced including:

Entity Integrity Constraints: is enforced through the use of primary keys. Each table contains a primary key attribute that uniquely identifies every record.

Please note that primary keys must be unique and cannot contain NULL values.

They are:

Customer(CustomerID)

KYC_Record(KYC_ID)

Branch(BranchID)

Account(AccountID)

Transaction(TransactionID)

Transaction_Log(LogID)

Card(CardID)

Loan(LoanID)

Employee(EmployeeID)

Referential Integrity Constraints: Referential integrity is used to enforce relationships between entities. These constraints prevent orphan records and guarantee that referenced records exist before dependent records are created.

- KYC_Record.CustomerID

  REFERENCES Customer(CustomerID)

- Account.CustomerID

  REFERENCES Customer(CustomerID)

- Account.BranchID

  REFERENCES Branch(BranchID)

- Transaction.AccountID

  REFERENCES Account(AccountID)
- Transaction_Log.TransactionID

  REFERENCES Transaction(TransactionID)
- Card.AccountID

  REFERENCES Account(AccountID)
- Loan.CustomerID

  REFERENCES Customer(CustomerID)
- Employee.BranchID

  REFERENCES Branch(BranchID)

Unique Constraints : To prevent duplication of critical banking information, the following attributes must contain unique values

Customer.Email

Customer.PhoneNumber

KYC_Record.BVN

Branch.BranchCode

Account.AccountNumber

Transaction.ReferenceNumber

Card.CardNumber

Employee.Email

Domain Constraints : Domain constraints restrict attribute values according to a rule.

Using Keyword CHECK Some are:

Customer

Gender IN ('M', 'F')

CustomerType IN ('Individual', 'Business')

Status IN ('Active', 'Inactive', 'Suspended')

KYC_Record

VerificationStatus IN ('Pending', 'Verified', 'Rejected')

DocumentType IN ('International Passport', 'National ID', 'Driver License', 'Voter Card')

Branch

BranchCode MUST be unique

PhoneNumber MUST contain valid numeric values

Account

CurrentBalance >= 0

AccountType IN ('Savings', 'Current', 'Fixed Deposit')

AccountStatus IN ('Active', 'Dormant', 'Closed')

Currency IN ('NGN', 'USD', 'GBP', 'EUR')

Transaction

Amount > 0

TransactionType IN ('Deposit', 'Withdrawal', 'Transfer')

TransactionStatus IN ('Pending', 'Completed', 'Failed')

Transaction_Log

ActionPerformed IN ('Created', 'Updated', 'Approved', 'Reversed', 'Cancelled')

LogTimestamp MUST contain a valid date and time

Card

CardType IN ('Debit', 'Credit', 'Prepaid')

CardStatus IN ('Active', 'Blocked', 'Expired')

ExpiryDate > IssueDate

Loan

LoanAmount > 0

InterestRate > 0

LoanTerm > 0

LoanStatus IN ('Pending', 'Approved', 'Rejected', 'Closed')

Employee

Department IN ('Operations', 'Customer Service', 'IT', 'Finance', 'Risk Management', 'Human Resources')

EmploymentDate <= Current Date

Email MUST follow a valid email format

NOT NULL Constraints: The following attributes are mandatory and cannot contain NULL values.

5.5 Mandatory Attribute Constraints (NOT NULL)

Customer

FirstName NOT NULL

LastName NOT NULL

DateOfBirth NOT NULL

PhoneNumber NOT NULL

Email NOT NULL

RegistrationDate NOT NULL

Status NOT NULL

KYC_Record

CustomerID NOT NULL

BVN NOT NULL

DocumentType NOT NULL

DocumentNumber NOT NULL

VerificationStatus NOT NULL

Branch

BranchName NOT NULL

BranchCode NOT NULL

StreetAddress NOT NULL

City NOT NULL

State NOT NULL

Country NOT NULL

PhoneNumber NOT NULL

Account

CustomerID NOT NULL

BranchID NOT NULL

AccountNumber NOT NULL

AccountType NOT NULL

CurrentBalance NOT NULL

Currency NOT NULL

DateOpened NOT NULL

AccountStatus NOT NULL

Transaction

AccountID NOT NULL

TransactionType NOT NULL

Amount NOT NULL

TransactionDate NOT NULL

ReferenceNumber NOT NULL

TransactionStatus NOT NULL

Transaction_Log

TransactionID NOT NULL

ActionPerformed NOT NULL

PerformedBy NOT NULL

LogTimestamp NOT NULL

Card

AccountID NOT NULL

CardNumber NOT NULL

CardType NOT NULL

IssueDate NOT NULL

ExpiryDate NOT NULL

CardStatus NOT NULL

Loan

CustomerID NOT NULL

LoanType NOT NULL

LoanAmount NOT NULL

InterestRate NOT NULL

LoanTerm NOT NULL

ApplicationDate NOT NULL

LoanStatus NOT NULL

Employee

BranchID NOT NULL

FirstName NOT NULL

LastName NOT NULL

JobTitle NOT NULL

Department NOT NULL

Email NOT NULL

PhoneNumber NOT NULL

EmploymentDate NOT NULL

Default Constraints: Default values are assigned where appropriate to improve data consistency

Customer.Status = 'Active'

Account.AccountStatus = 'Active'

Transaction.TransactionStatus = 'Pending'

Card.CardStatus = 'Active'

KYC_Record.VerificationStatus = 'Pending'

RegistrationDate = Current Date

DateOpened = Current Date

TransactionDate = Current Timestamp

LogTimestamp = Current Timestamp

Business Rule Constraints

The below are business rules that Apex Digital Bank database adheres to:

1. A customer may own multiple accounts.
2. Each account belongs to one customer.
3. Each account is managed by one branch.
4. A branch may manage multiple accounts.
5. An account may have multiple transactions.
6. Each transaction belongs to one account.
7. Each transaction may generate multiple audit log records.
8. A customer may apply for multiple loans.
9. A customer can possess only one KYC record.
10. An account may be linked to multiple debit or credit cards.

11. A branch may employ multiple employees.
12. Each employee is assigned to one branch.

Database Normalization

What is Normalization in Databases?

Database normalization is the process of organizing data into multiple tables to minimize redundancy, eliminate undesirable dependencies, and improve data integrity. The Apex Digital Bank database was designed and normalized to ensure efficient storage, consistency, and maintainability of banking data.

First Normal Form (1NF):
The database satisfies 1NF because each table contains a primary key, all attributes store atomic (single) values, and there are no repeating groups or multivalued attributes.

Second Normal Form (2NF):
The database satisfies 2NF because all non-key attributes are fully dependent on their respective primary keys. Since each table uses a single-attribute primary key, no partial dependencies exist.

Third Normal Form (3NF):
The database satisfies 3NF because there are no transitive dependencies. Customer, Branch, Account, Transaction, Loan, Card, KYC, and Employee data are stored in separate tables, ensuring that non-key attributes depend only on the primary key of their respective table.